University of Mosul
College of Computer Sciences and Mathematics
Computer Science Dept. /First Class

Principles of Computer Organization
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presentation
notes

Lec. 6

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80X86 Addressing Modes

The general format for an assembly language statement is

LABLE: INSTRUCTION ; COMMENT

An example of a complete assembly language statement is

Start: MOV AX,CX ; Copy CX into AX

address identifier (label)	Instruction	Comment

Operation Codes and Operand

Instructions are made up of two parts:

1. Operation Code or Op-Code specifies the operation to be performed.
2. Operand identify source and destination of the data acted on by the op-code.

Example:

MOV AX,BX

op-code

destination operand

source operand

80X86 Addressing Modes

Addressing Mode: is the way in which MP can access operands.
80x86 provide a total of seven addressing modes:

1. Register
2. Immediate
3. Direct
4. Register Indirect
    a. Based.
    b. Index.
    c. Based Index.
    d. Base with Displacement.
    e. Index with Displacement.
    f. Base Index with Displacement.
5. Stack addressing mode.
6. String addressing mode.
7. Input output addressing mode.

1. Register addressing mode

- use of registers to hold the data to be manipulated
- Memory is not accessed when this addressing mode is executed
- therefore, it is relatively fast
- Example:

```
MOV   BX, DX   ;copy the contents of DX into BX
MOV   ES, AX   ;copy the contents of AX into ES
ADD   AL, BH   ;add the contents of BH to contents of AL
```

- Source and destination registers must have the same size

```
Example      DL=6F   , SI = 4000
    MOV      CL,DL
    MOV      BX,SI
```

CL= 6F
BX= 4000

2. Immediate addressing mode

- the source operand is a constant
- the operand comes immediately after the opcode
- For this reason, this addressing mode executes quickly
- Immediate addressing mode can be used to load information into any of the registers except the segment registers and flag registers.
- Ex:
  - MOV AX, 2550H ;move 2550H into AX
  - MOV CX, 625 ;load the decimal value 625 into CX
  - MOV BL, 40H ;load 40H into BL

Example 1

MOV AL,15

AL= 15

EX1:

```
MOV    SI,F902
MOV    BX,SI
MOV    AL,BL
MOV    DX,3F2
```

بعد التنفيذ الايعاز الاول

SI=F902

بعد تنفيذ الايعاز الثاني

BX=F902

بعد تنفيذ الايعاز الثالث

AL=02

بعد تنفيذ الايعاز الرابع

DX=03F2

2. Immediate addressing mode

☞ ***Moving information to the segment registers***

- the data must first be moved to a general-purpose register and then to the segment register.
- Example:
- MOV AX, 2550H
- MOV DS, AX
- ~~MOV DS, 0123H~~ ;illegal instruction!

Ex1: Define data segment at address 9000

الحل :

MOV DX,9000
MOV DS,DX

3. Direct addressing mode

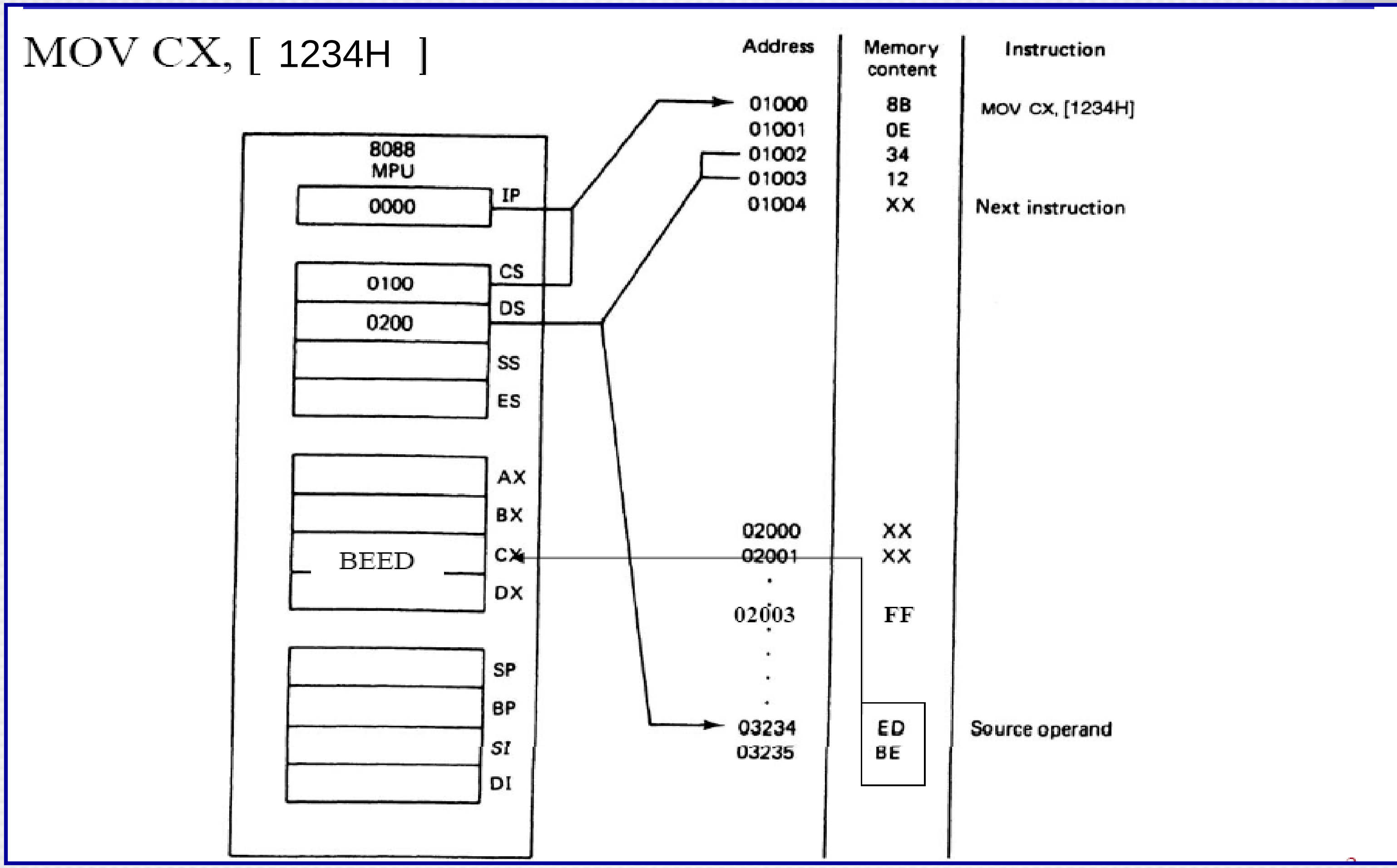
- the data is in some memory location(s) and the address of the data in memory comes immediately after the instruction
- This address is the offset address
- Example
  - MOV DL, [2400] ;move contents of DS:2400H into DL
- the physical address is calculated by combining the contents of offset location 2400 with DS

Note the brackets

PA= segment Base : Direct offset address

$$PA = \begin{Bmatrix} CS \\ DS \\ SS \\ ES \end{Bmatrix} : \{Direct\ offset\ address\}$$

Example



MOV CX, [1234H]
Address
Memory content
Instruction
8088 MPU
IP
0000
CS
0100
DS
0200
SS
ES
AX
BX
CX
BEED
DX
SP
BP
SI
DI
01000 8B MOV CX, [1234H]
01001 0E
01002 34
01003 12
01004 XX Next instruction
02000 XX
02001 XX
02003 FF
03234 ED Source operand
03235 BE

Examples

Example1:

Find the physical address of the memory location and its contents after the execution of the following, assuming that DS = 1512H.

```
MOV    AL, 99H
MOV    [3518], AL
```

MOV AL, 99H → Immediate addressing mode
MOV [3518], AL → Direct addressing mode

الحل :

AL= 99   بعد تنفيذ الايعاز الاول

الايعاز الثاني

PA= SEG.* 10 + OFFSET

= 1512*10 + 3518= 15120+3518=18638

	:
18638	99
	:

Examples

Example2: ASSUME THAT DS= F900

MOV DX, F205 → Immediate addressing mode

MOV [2000],DX → Direct addressing mode

MOV BH,[2000] → Direct addressing mode

الحل :

تنفيذ الايعاز الثاني :

Ph= DS*10 + OFFSET

PH= F9000+ 2000=FB000

الايعاز الثالث التنفيذ

(DS:2000) → BH

BH= 05

LOGICAL ADDRESS
DS:2000 → FB000

Address	Content
	:
FB000	05
FB001	F2
	:

H.W

If you have the following instructions, and assume that DS=F042.

```
MOV   AX,66FF
MOV   DI,0022
MOV   [4000],Ax
MOV   BL,[4001]
MOV   AH,BL
```

Answer the Following Question:

Identify the addressing mode for above instructions?

حدد نوع Addressing mode

Find the **physical address** of memory locations **and its contents** after execution of above instructions.

احسب physical address لموقع الذاكرة بالايعاز الثالث و ماهي محتويات الموقع

What is the content of AX register? ماهي محتويات

What is the content of BL register?